

Data_Visualization_Report

Code Alpha Data Analytics Internship

Task 3: Data Visualization using Superstore Dataset

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1. Introduction

Data Visualization is the process of presenting raw data in graphical formats such as charts, graphs, and dashboards. It helps users understand patterns, trends, relationships, and insights that may not be obvious from tables of data. In this project, the Superstore dataset was analyzed to evaluate sales performance, profitability, customer segments, and regional trends using Python visualization libraries.

2. Dataset Description

- **Dataset Name:** Superstore Dataset
- **Number of Records:** 9,994
- **Number of Columns:** 21

The dataset contains sales transactions, customer information, shipping details, product categories, regions, and profitability data from a retail store.

3. Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- VS Code

4. Data Preparation

The following preprocessing steps were performed:

- Loaded the dataset into Pandas.
- Explored the dataset structure.
- Checked column names and data types.
- Verified missing values.
- Removed duplicate records.
- Converted date columns into date time format.

5. Visualizations

The following visualizations were created:

1. Sales by Category
2. Profit by Category
3. Sales by Region
4. Profit by Region
5. Sales by Customer Segment
6. Monthly Sales Trend
7. Top 10 States by Sales
8. Top 10 Cities by Sales
9. Discount vs Profit
10. Correlation Heatmap

For each visualization, include:

- Chart image
- Observation
- Business Insight

6. Business Insights

Key findings from the analysis include:

- Technology products generated the highest sales.
- Consumer customers were the largest customer segment.
- The West region achieved high sales performance.
- High discounts frequently reduced profit.
- Monthly sales showed fluctuations, indicating seasonal demand.

7. Data Storytelling

The visualizations reveal how product categories, customer segments, regions, and pricing strategies affect overall business performance. While sales are strong in certain regions and product categories, excessive discounts can reduce profitability. These insights can help managers improve pricing, inventory management, and marketing strategies.

8. Conclusion

This project successfully transformed raw sales data into meaningful visualizations using Python. The analysis identified trends, customer behavior, regional performance, and profitability, supporting better business decisions. The project fulfills the requirements of **CodeAlpha Task 3: Data Visualization** and demonstrates practical data visualization skills.

9. References

- Superstore Dataset
- Python Documentation
- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation